

AnchorPro Construction

Scope of Work, Requirements & Development Roadmap

Pivoting the AnchorPro Platform to a Construction-Focused ERP

Prepared: 24 August 2026

Basis: contractor/site-manager workflow requirements and prior codebase audit

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1. Introduction & Background

AnchorPro began as a multi-tenant ERP/field-service platform for maintenance, workshop, and asset-based operations. Following an independent codebase and product audit, and on the recommendation of an Information Systems / software engineering consultant, the development team elected to pivot the product toward a single, well-defined vertical: construction site and contract management for small-to-mid-sized contractors.

The scope in this document is built directly around a first-hand workflow account from a working Construction Management practitioner, rather than generic industry literature. That account described three linked layers of work: (1) daily/weekly/monthly site reporting, (2) quantity-surveying documents — the Bill of Quantities (BOQ) and Interim Payment/Progress Certificates — and (3) a surrounding toolchain of Microsoft Project for scheduling and Autodesk Construction Cloud for drawings.

Note on billing model: this platform is being built around the Bill of Quantities / Interim Payment Certificate model used across the Commonwealth and Southern African construction industry (JBCC/NEC/FIDIC-family contracts), not the AIA G702/G703 pay-application model used in the United States. The underlying goals are similar — measure completed work, apply agreed rates, deduct retention, get certified for payment — but the documents, terminology, and calculation logic differ, and the system should be built around BOQ line items and measured value, not a Schedule of Values.

2. Scope of Work

This phase of work covers the redesign and extension of the AnchorPro platform from its current field-service data model into a construction-project data model, and the design and build of the modules described in Section 3.

2.1 In Scope

- Redesign of the core data model from job-card-centric to project/contract-centric, with Bills of Quantities as first-class structured data.
- A mobile-first Site Diary module for daily data capture (labor, plant, deliveries, weather, safety, photos).
- Automated weekly and monthly report generation from site diary and schedule data, replacing the current manual Word/PDF process.
- A Bill of Quantities (BOQ) module supporting takeoff-based quantities, supplier/subcontractor rates, and trade/section groupings.
- A Progress / Interim Payment Certificate module with measurement against BOQ items, variations, retention, and an approval workflow.
- A Variations (change order) module valued against BOQ rates.
- Baseline-schedule import from Microsoft Project and progress-against-program tracking.
- Reference-level linking of drawings and revisions from Autodesk Construction Cloud against BOQ items and site diary entries.

- Retention of and continued build-out for existing carried-over modules: multi-tenancy, authentication/roles, finance/ledger, HR/payroll, procurement/inventory, safety/permits, dashboards/BI, and the Copilot assistant layer.
- Remediation of the security and code-quality findings raised in the prior codebase audit, as a precondition for any client-facing pilot.

2.2 Out of Scope (this phase)

- Full replacement of Microsoft Project (scheduling engine) or Autodesk Construction Cloud (drawing management/markup) — both are integrated with, not rebuilt.
- BIM model viewing or 3D coordination.
- Public-tender / bid management workflows.
- Certified payroll / prevailing-wage calculations for jurisdictions that require them, unless a specific pilot client needs it.
- Accounting-system-of-record functionality (the finance module supports project cost control and certificates; it is not positioned to replace a statutory accounting package).

3. Functional Requirements

Requirements are grouped by module and tagged with a MoSCoW priority: Must (required for a usable pilot), Should (expected soon after), Could (valuable, deferrable).

3.1 Site Diary

ID	Requirement	Priority
FR-1.1	User can create one site diary entry per project per day, capturing date, weather conditions, and free-text notes on work performed.	Must
FR-1.2	User can record labor headcount on site, ideally broken down by trade or crew.	Must
FR-1.3	User can record plant/equipment present on site and hours used, linked to the existing equipment register.	Must
FR-1.4	User can log deliveries received, including supplier, material, and quantity.	Must
FR-1.5	User can log safety issues or incidents, linked to the existing safety/permit module.	Must
FR-1.6	User can attach one or more photos to a diary entry from a mobile device.	Must
FR-1.7	Diary entries can be created and saved offline, then synced when connectivity is available.	Should

3.2 Weekly Reporting

ID	Requirement	Priority
FR-2.1	System automatically aggregates the current week's diary entries into a draft weekly progress summary.	Must

FR-2.2	System compares actual progress recorded during the week against the imported schedule baseline and flags variances.	Should
FR-2.3	User can add or edit narrative commentary (issues to flag, plan for next week) before issuing the report.	Must
FR-2.4	Weekly report can be exported as PDF.	Should

3.3 Monthly Client/Consultant Reporting

ID	Requirement	Priority
FR-3.1	System generates a draft monthly report combining: overall progress percentage, cost-to-date summary, aggregated safety statistics, and a rolled-up narrative of challenges from weekly reports.	Must
FR-3.2	User (Director or delegate) can review and edit the draft report before approval.	Must
FR-3.3	Approved monthly report is exported as a branded, print-ready PDF matching the firm's existing template style.	Must
FR-3.4	System retains a version history of issued monthly reports per project.	Should

3.4 Schedule Integration

ID	Requirement	Priority
FR-4.1	User can import a baseline schedule exported from Microsoft Project (e.g., XML) against a project.	Must
FR-4.2	System displays planned vs. actual progress at task/milestone level.	Should
FR-4.3	User can re-import an updated schedule baseline when the program is revised.	Could

3.5 Drawing Reference

ID	Requirement	Priority
FR-5.1	User can attach a reference link and revision number for drawings held in Autodesk Construction Cloud to a project, BOQ section, or diary entry.	Should
FR-5.2	System flags when a linked drawing reference has not been revisited since a newer revision was noted.	Could

3.6 Bill of Quantities (BOQ)

ID	Requirement	Priority
FR-6.1	User can create a BOQ for a project, structured into trade/work sections.	Must
FR-6.2	Each BOQ line item captures a description, unit of measure, quantity, rate, and calculated total.	Must

FR-6.3	User can import BOQ line items in bulk from a spreadsheet to ease migration from existing Excel-based BOQs.	Must
FR-6.4	System calculates and displays the total contract sum from all BOQ line items.	Must
FR-6.5	BOQ line items can be revised through a controlled version history once a contract is signed.	Should

3.7 Progress / Interim Payment Certificates

ID	Requirement	Priority
FR-7.1	User can create a new certificate for a project for a given period, pre-populated with BOQ line items.	Must
FR-7.2	For each BOQ line, user can record quantity/value completed this period and the system calculates cumulative value to date and percentage complete.	Must
FR-7.3	Approved variations can be added to a certificate and valued against BOQ rates or a manually agreed rate.	Must
FR-7.4	System calculates retention deduction based on a configurable percentage and contract-specific rules (including step-down/release milestones).	Must
FR-7.5	System calculates net amount due after deducting retention and previous certificates paid.	Must
FR-7.6	Certificate can attach supporting photo evidence, pulled from linked site diary entries.	Should
FR-7.7	Certificate supports a review/approval workflow with distinct states: Draft, Submitted to Consultant, Queried, Approved, Issued, Paid.	Must
FR-7.8	Certificate can be exported as a PDF containing a summary sheet and a detailed line-item breakdown.	Must

3.8 Variations

ID	Requirement	Priority
FR-8.1	User can raise a variation against a project, referencing affected BOQ items or adding new priced items.	Must
FR-8.2	Variations carry an approval status and, once approved, automatically become available for inclusion in the next certificate.	Must
FR-8.3	System maintains an audit trail of who raised, reviewed, and approved each variation.	Must

3.9 Carried-Over Modules (from the existing AnchorPro platform)

ID	Requirement	Priority
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FR-9.1	Multi-tenant account structure, authentication, and role-based permissions continue to apply to construction-specific modules.	Must
FR-9.2	Existing procurement/inventory functionality is adapted so materials are tracked against a project and, where relevant, a BOQ line.	Should
FR-9.3	Existing HR/payroll functionality continues to support site labor records feeding into diary headcount and cost tracking.	Should
FR-9.4	Existing finance/ledger functionality is extended to reconcile against certificates issued and paid.	Must
FR-9.5	Existing safety/permit module feeds safety statistics into weekly and monthly reports.	Must
FR-9.6	Existing dashboard/BI layer is extended with project-level cost, progress, and certificate status views.	Should
FR-9.7	Existing Copilot assistant layer can answer natural-language questions about a project's progress, cost, and certificate status.	Could

4. Non-Functional Requirements

Category	Requirement
Security	All findings rated Critical or High in the prior codebase audit (hardcoded credentials, exposed password-reset token, tenant-isolation edge cases) must be remediated before any client-facing pilot. Financial data — certificates, retention, variations — requires role-gated access and a full audit trail of who changed what and when.
Multi-tenancy & data isolation	Tenant data isolation must hold under the new project/BOQ/certificate data model with the same rigor as the existing platform; every new table carries a tenant identifier and is covered by the global query filter.
Offline / field usability	Site diary entry must work in low- or no-connectivity conditions common on construction sites, with reliable sync once connectivity returns and no silent data loss.
Performance	Report and certificate PDF generation should complete within a few seconds for a typical project; dashboards should load without noticeable delay for a portfolio of active projects per tenant.
Usability	Site-level data entry (diary, photos) must be usable one-handed on a mobile device by non-technical site staff; office-side BOQ and certificate screens can assume a QS/accounts user on a desktop.
Data integrity & auditability	Once a certificate is submitted for approval, its figures must be immutable except through a tracked revision, given its role as a financial and potentially legal document.
Interoperability	The system must accept schedule exports from Microsoft Project and reference links from Autodesk Construction Cloud; certificates and reports must export to PDF, and BOQs must both import from and export to spreadsheet formats.

Localization	Currency, unit of measure (metric), and date formatting must suit the target Southern African market; the system should support multiple currencies where a tenant operates across borders.
Scalability	The data model should support a contractor running multiple concurrent projects per tenant without degradation, since this is the normal operating mode for the target customer segment.
Reliability & backup	Certificate and BOQ data, once approved, must be durably backed up; loss of a certified payment record has direct financial and legal consequences for the customer.
Maintainability	New construction modules should follow the same architectural conventions (EF Core, tenant filtering, service/controller separation) as the existing platform to keep the codebase coherent for a small team.

5. Development Roadmap

The roadmap below assumes a small development team working iteratively, with each phase producing something demonstrable. Durations are planning estimates for sizing purposes, not fixed commitments.

Phase	Focus	Key Deliverables	Est. Duration
Phase 0	Discovery & Data Model Design	Confirm contract form(s) in use (JBCC, NEC, FIDIC, or bespoke), standard method of measurement, and retention rules with the Construction Management contact. Redesign core schema from job-card-centric to project/BOQ-centric.	2 – 3 weeks
Phase 1	Core Foundations	Project entity and BOQ module (FR-6.x); carry over and adapt tenant, auth, and finance base to the new model.	4 – 6 weeks
Phase 2	Field Data Capture	Mobile Site Diary module (FR-1.x) with offline sync; linkage to existing equipment, procurement, and safety modules.	3 – 5 weeks
Phase 3	Reporting Engine	Automated weekly rollups (FR-2.x) and monthly report generation (FR-3.x); Microsoft Project baseline import (FR-4.x).	3 – 4 weeks
Phase 4	Certification & Billing	Progress/Interim Payment Certificate module (FR-7.x), Variations module (FR-8.x), retention engine, approval workflow.	4 – 6 weeks
Phase 5	Integrations & Field Polish	Drawing reference linking to Autodesk Construction Cloud (FR-5.x); dashboard/Bi and Copilot extensions (FR-9.6, FR-9.7).	3 – 4 weeks
Phase 6	Security & Quality Hardening	Remediate Critical/High findings from the prior audit; expand automated test coverage beyond API smoke tests; resolve outstanding build warnings.	2 – 3 weeks
Phase 7	Pilot & UAT	Run a live pilot on one active project with the Construction Management contact; collect feedback	2 – 3 weeks

		on the diary, reports, and certificate workflow; fix and refine.	
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Indicative total: roughly 23 – 34 weeks (5.5 – 8 months) from discovery to a pilot-ready system for a small team, before wider rollout.

6. Assumptions & Dependencies

- The Construction Management contact (or an equivalent site-level user) remains available for validation throughout Phases 0, 2, 3, and 7.
- Microsoft Project files or exports can be obtained from client sites for schedule import testing.
- Autodesk Construction Cloud access (or sample exports) is available to validate drawing reference linking; full API integration depends on account tier and API access.
- The exact contract form and standard method of measurement in use will be confirmed during Phase 0, as this materially affects the BOQ and certificate data structures.
- Existing AnchorPro tenant, auth, and finance modules are stable enough to extend rather than rebuild; this roadmap assumes extension, not replacement.

7. Risks Carried Over From Prior Audit

The prior codebase audit identified several findings that apply directly to this build, since certificates and BOQs are financial and potentially legal documents:

- Hardcoded credentials and secrets in seed data and configuration (Critical) — must be resolved before any pilot handles real client data.
- Password reset token exposed in API response (Critical) — a direct account-takeover risk once contractors, consultants, and clients are all authenticating into the same system.
- Multi-tenant filter edge cases (High) — must be re-audited against the new project/BOQ/certificate tables, since cross-tenant leakage of certificate data would be a serious incident for any customer.
- MailKit dependency vulnerability (Medium) and outstanding nullable-reference build warnings (Low) — should be cleared during Phase 6 rather than carried indefinitely.

8. Glossary

Term	Definition
BOQ (Bill of Quantities)	A structured list of project work items with measured quantities, units, and priced rates; forms the basis of the contract sum and later valuations.
Progress / Interim Payment Certificate	A periodic (typically monthly) document certifying the value of work completed to date, used to authorize payment to the contractor.
Retention	A percentage of certified value withheld by the client as security until defined completion milestones are met, then released in stages.

Variation	An approved change to the contracted scope, valued against BOQ rates or a newly agreed rate, and incorporated into subsequent certificates.
QS (Quantity Surveyor)	The professional (often representing the client) who measures completed work and validates certificate valuations.
Program	The construction industry term for the project schedule, typically maintained in Microsoft Project.
Site Diary	A daily record kept by the site manager of labor, plant, deliveries, weather, and safety events.